

LESSON 111 (0.7 GROUND and 1.3 DUAL)

Lesson Objective

During this lesson, the student will be introduced to flight at minimum controllable airspeed. The student will learn the signs of an incipient and full stall and the differences between them. The student will also learn the skills and risk management associated with power off stalls and stall prevention. Also, the student will be introduced to the rectangular course.

Briefing

- Outline the objectives and completion standards of the lesson.
- Establish the student and instructor responsibilities.
- Review homework assigned in the previous lesson.
 - Quiz the student on the readings as needed.
 - Chapter 5 of the Airplane Flying Handbook - Maintaining Aircraft Control: Upset Prevention and Recovery Training.
 - Ask the students if they have any questions on the procedures from the maneuver's manual.
 - Power off stalls.
 - Maneuvering during slow flight.
- Student questions.

References

- Advisory Circular. ([Search Tool](#))
 - Stall and Spin Awareness Training. ([AC 61-67](#))
- Airplane Flying Handbook. ([FAA-H-8083-3C](#))
 - Chapter 05 - Maintaining Aircraft Control: Upset Prevention and Recovery Training.
 - Chapter 07 - Ground Reference Maneuvers.
- Warrior Maneuvers Manual.

Lesson Content (Ground)

Power-Off Stall

- Power-off stalls occur at low-powered conditions during approach to landing.
- Explain the difference between the incipient stage and the fully developed stage.
- Explain how stall speed is directly affected by load factor and bank angle.
 - Load factor = $1 / \cos(\text{angle of bank})$.
 - Stall speed banked = stall speed level x square root of load factor.
- It is important for the student to understand stall characteristics as well as indications of a stall in order to provide the proper recovery procedures.
 - Indications:
 - Stall warning.
 - Buffeting.

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- Slow airspeed.
- Reduced control effectiveness.
- Recovery procedures must be prompt and positive in order to achieve minimal loss of altitude.
- Quiz the student on the procedures for a power-off stall, as this should have been reviewed for homework.
- Answer any questions or doubts.

Maneuvering During Slow Flight

- Maneuvering during slow flight will help the students to develop a better feel for how the aircraft handles at a low airspeed.
- This should lead to a positive transfer of learning into controlling the aircraft during approach and landing.
- Under the conditions assumed in slow flight, the region of reverse command is entered.
 - This happens in the “back half” of the power curve. (draw this for student)
 - Here power controls altitude and pitch controls airspeed.

Verify No Passengers

- This lesson should not include any observers.

Go/No-Go Decision

- The student must make a go/no-go decision depending on:
 - Personal minimums.
 - FIT minimums.
 - Aircraft.
 - Airspace.
- Review the students planning with a focus on:
 - Weather, weight and balance, performance, flight planning, and fuel calculations.
 - NOTAMs, TFRs, FIFs, aircraft documents, and personal documents.
 - Equipment requirements (e.g. flashlight, fuel card, drinking water, view-limiting devices, life jackets, fly away kit, survival kit, etc.)
 - FRAT/TEM/safety considerations for the flight.
- While this is represented by one line item prior to departure, the go/ no-go decision is continuous and should be a consideration even once airborne.

Lesson Content (Dual)**Preflight Inspection**

- Ensure that the student conducted a thorough preflight inspection.

Visual Scanning and Collision Avoidance

- Emphasize that it is the pilot in command's responsibility to avoid traffic.
- IF YOU SEE SOMETHING, SAY SOMETHING!

Normal/Crosswind Takeoff

- The student should conduct the takeoff.
- Ensure the runway entry check and the lineup check are completed.
- Ensure proper crosswind correction is maintained.

Review Previous Exercises as Required

- Use the appropriate lesson plan depending on what needs to be reviewed.

Maneuvering During Slow Flight and Set Up for the Stall

- Instructor does, student says:
 - Verify correct procedure.
 - Stress smoothness and coordination.
 - Demonstrate how to promptly adjust for being too low, too high, too fast, and too slow.
 - Demonstrate turns, climbs, and descents.
- Student does, student says:
 - Verify correct procedure.
 - Stress smoothness and coordination.
 - Coach the student as needed.

Symptoms of the Approaching Stall

- As the airplane approaches the stall, point out the indications of the approaching stall.
 - Decreasing airspeed.
 - Higher than normal angle of attack.
 - Reduced control effectiveness.
 - Stall warning.
 - Light buffeting.

Symptoms of the Full Stall

- As the airplane enters the full stall stage, point out the indications.
 - Heavy buffeting.
 - Nose drop.
 - High rate of descent.
 - If the airplane is not coordinated → wing drop.

Fully Developed Stall Recovery Without Power

- For the first stall, bring the airplane to a full stall and recover without using power.
- Note the altitude loss.

Fully Developed Stall Recovery With Power

- For the second stall, bring the airplane to a full stall and recover using power.
- Note the altitude loss.

Enhanced Private Pilot Certification Course**Incipient Stall Recovery**

- For the third stall, recover from the first indication.
- Note the altitude loss.
- Compare the altitude loss difference between the three different recoveries.
 - Emphasize to the student that this is why it is important to be aware of early indications of a stall.

Rectangular Course

- Help the student find 4 points creating a large rectangle.
- Enter the downwind leg at a 45-degree angle and 2500 ft.
- Practice non-specific (base) vs specific (final) aiming point techniques.
- Initiate a go-around on the final leg at 1500 ft.

Normal Traffic Pattern

- Provide guidance for the student through the legs of the pattern.
 - Power settings.
 - Airspeeds.
 - Aiming points.
 - Crosswind correction.

Normal/Crosswind Landing

- Instructor conducts the landing.
- Provide guidance for the student through the procedure, control inputs, and visual cues.
- Have the student follow along on the controls to gain a deeper understanding of what the instructor is doing.
- Have the student pay attention to the sight picture, and placement of the aiming point on final.

Go-Around/Rejected Landing

- After conducting the rectangular course, have the student conduct a go-around.
- Provide guidance for the student through the go-around procedures.
 - Emphasize the importance of not going directly to flaps 0
 - Emphasize the importance of adding power first.
- Ensure that the student uses TRIM while climbing out.

Postflight Procedures

- Ensure that the student conducts a thorough postflight inspection.

Post-Brief

- Student self-critique.
- Instructor critique.
- Questions from the student.
- Lesson grading.
- Logbook completion.

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- Schedule the next lesson.
- Preview the next lesson → 112.
- Assign homework:
 - Review the Warrior Maneuvers Manual:
 - Power-off stall.
 - Power-on stall.
 - Turns around a point.
 - Review the Airplane Flying Handbook:
 - Chapter 07 (ground reference maneuvers).

Completion Standards

This lesson is complete when the student is able to:

1. Correctly perform previously learned maneuvers with little instructor guidance.
2. Demonstrate proper sequential checklist procedures for all normal procedures without instructor guidance.
3. Demonstrate flight at minimum controllable airspeed with instructor guidance.
4. Recognize the difference between incipient and fully developed stalls.
5. Recover from incipient and fully developed stalls with instructor guidance.
6. Conduct traffic patterns, takeoffs and landings with instructor guidance.